Understanding AfterSight from scratch

A beginner's guide to every idea, file and number in this project, written for viva preparation.

You need basic Python and the very basics of machine learning. Everything else is explained the first time it appears. Every number in this guide comes from this repository: saved results, the cached data, or a script that traced a real building through the real code (`docs/build/understanding_data.py`).

Contents

1. The big picture
2. The basics you need first
3. Machine learning basics, as used here
4. Follow one real building end to end
5. Each pipeline stage explained
6. The algorithms, intuitively
7. Understanding the results
8. How the app works
9. Honest limitations
10. Code map
11. Glossary
12. Test yourself (answers at the very end)

1. The big picture

The problem. After a hurricane, flood, wildfire or tsunami, rescue and relief teams need to know quickly **which buildings are damaged and how badly**. Walking street by street takes days and can be dangerous. Satellites photograph whole cities from above, both before and after the disaster, but nobody can inspect tens of thousands of roofs by eye.

What AfterSight does. For every building in a satellite scene it predicts one of four damage levels: **no damage, minor damage, major damage or destroyed**. It also shows **why** the model made that prediction.

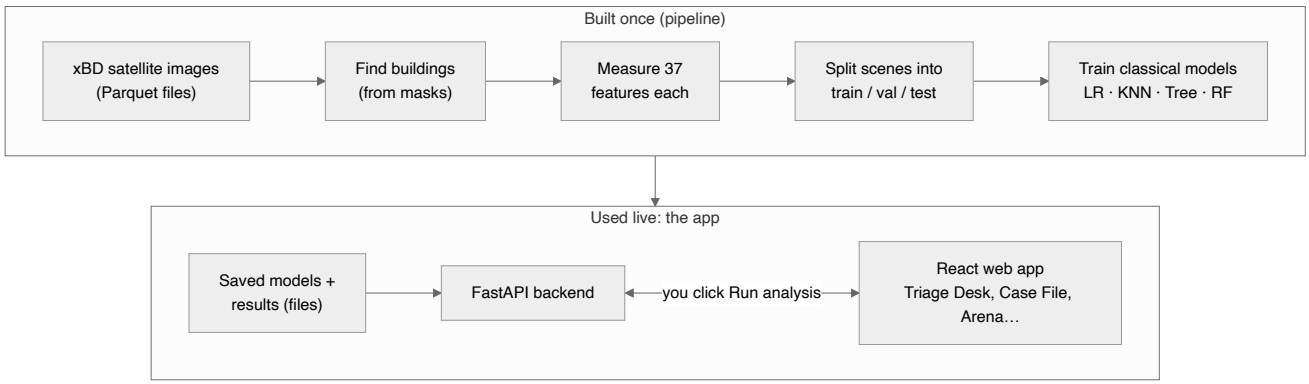
Who would use it. In real life it would be a damage analyst at a relief agency, as a first pass before sending people to check. In this project the users are us and our examiners: AfterSight is a research and teaching tool, not an operational system.

What goes in and what comes out.

In	Out
a *pre-disaster* satellite image and a *post-disaster* image of the same place, plus the outline of every building	for each building: a damage class, a confidence score, an explanation, and a flag if a human should double-check it

An analogy. Think of an insurance assessor with two photos of a street, one from last year and one from today. For each house they compare the photos ("the roof was smooth grey, now it is a rough pile of rubble"), decide a damage level, and write down their reasons. AfterSight does the same with numbers: it measures colour, roughness and change for each house, and a classical machine-learning model turns those measurements into a damage level.

The whole system on one page:



The one result to remember. On *new scenes from disasters the models were trained on*, the default model scores about **0.53** (macro-F1, explained in Section 7). On a disaster it has *never seen*, it scores about **0.29**. Most of this guide builds up to understanding that gap.

2. The basics you need first

2.1 What a satellite image is

A digital image is a grid of tiny squares called **pixels**. Each pixel stores three numbers between 0 and 255: how much **red**, **green** and **blue** light it has. These three layers are the image's **channels** (RGB). 0 means none of that colour and 255 means the maximum. A grey roof has roughly equal R, G and B values; grass has more green.

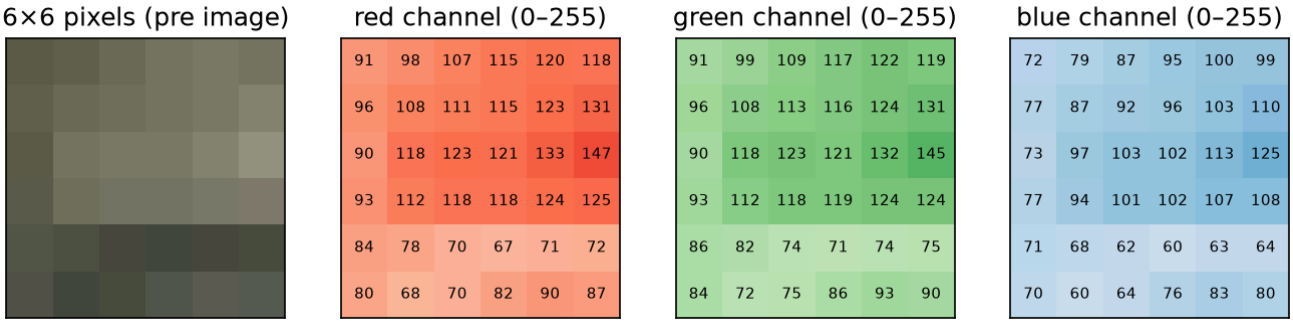


Figure 1: A 6×6-pixel corner of a real roof (the building we follow in Section 4). The top-left pixel is R = 91, G = 91, B = 72.

Every xBD image in this project is **1024 × 1024 pixels**, about a million pixels per image. **Resolution** says how much ground one pixel covers. The xBD authors chose imagery finer than **0.8 metres per pixel** [1]. At that scale a house is a few dozen pixels across, and fine details (a few missing roof tiles) are close to invisible.

2.2 Pre-disaster and post-disaster images

For each place, xBD gives two photos: **pre** (taken before the disaster) and **post** (taken after). Damage shows up as *change* between the two. A building that burned down looks dark and smooth before and grey-white and patchy after. Most of AfterSight's measurements compare pre and post.

2.3 What a mask is

A **mask** is an image of the same size where each pixel holds a *label* instead of a colour. Our data has two:

- the **footprint mask** (`t1_mask`): 1 where a building is, 0 elsewhere;
- the **damage mask** (`t2_mask`): 0 for background, then **1 = no damage, 2 = minor, 3 = major, 4 = destroyed**.

So the damage mask says, pixel by pixel, how damaged the building under that pixel is.

2.4 What xBD is and why it was made

xBD is a public dataset created for the **xView2** challenge (Gupta et al., 2019 [1]). People annotated building outlines on pre/post satellite pairs from many disasters (19 in the full dataset). Each building was graded on the four-level damage scale. It was made because no large, standard dataset existed for training and comparing damage-assessment methods. It is licensed **CC BY-NC-SA 4.0** (free for non-commercial use with credit).

Our copy is a **Parquet mirror** of xBD's training split: 17 files, about 8 GB, **2,799 scene pairs** from 10 disasters. **Parquet** is a compressed table file format; here each row holds one scene's two images and two masks. We use **5 disasters** (Table 1). They were chosen because they contain enough major and destroyed buildings; some others had almost none.

Event	Disaster type	Scenes	Buildings	No damage	Minor	Major	Destroyed
hurricane-harvey	flooding	319	19,121	9,463	2,251	7,034	373
hurricane-matthew	wind	238	9,570	2,103	4,440	1,140	1,887
hurricane-michael	wind	343	21,429	13,870	4,998	1,841	720
palu-tsunami	tsunami	113	9,157	7,360	1	200	1,596
santa-rosa-wildfire	fire	226	12,063	8,653	76	55	3,279
Total		**1,239**	**71,340**	**41,449**	**11,766**	**10,270**	**7,855**

Table 1: The five events used (from the inspection report and the feature table). "Scenes" counts every cached scene; 60 of them contain no labelled building, so the splits below cover 1,179 scenes.

2.5 What the four damage classes look like from above

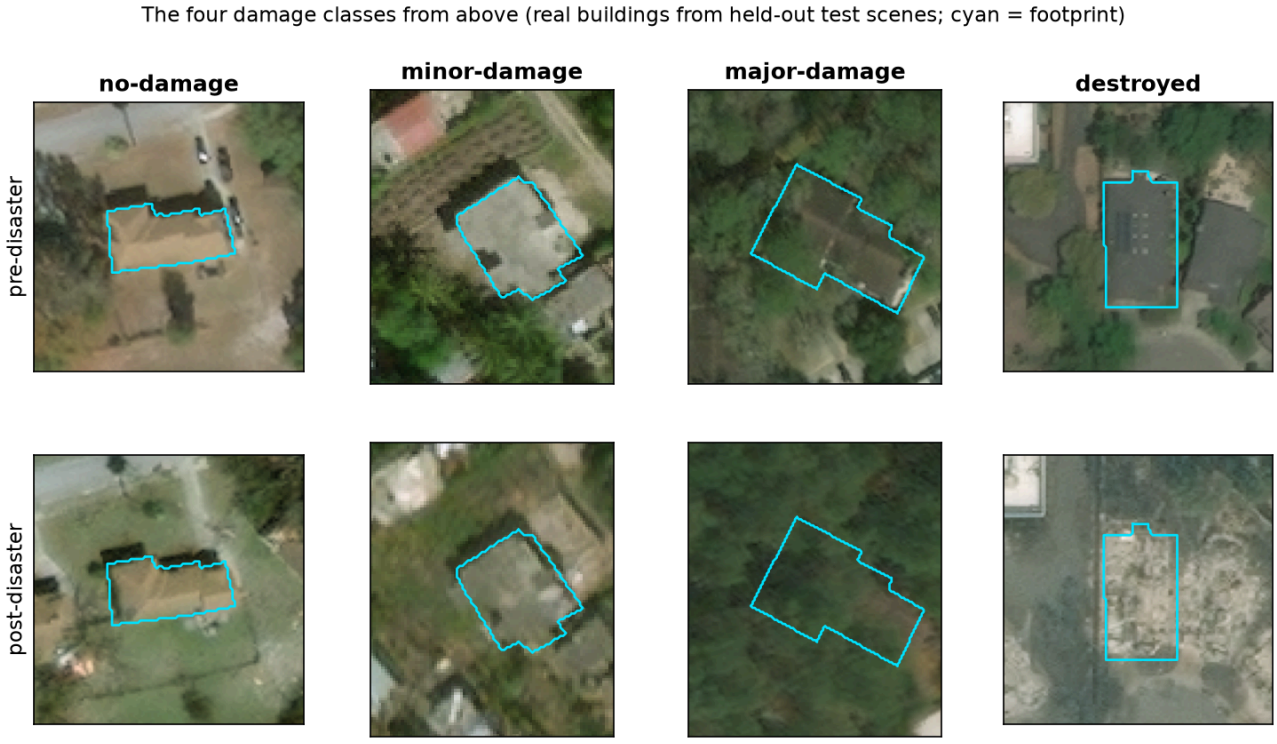


Figure 2: One real building per class from held-out test scenes (cyan = footprint): `hurricane-michael_00000009_b0008` (no damage), `hurricane-matthew_00000060_b0016` (minor), `hurricane-harvey_00000029_b0041` (major), `santa-rosa-wildfire_00000006_b0002` (destroyed).

Look closely. *No damage* and *destroyed* are obvious. *Minor* and *major* are hard to tell apart even for a person: the major-damage house is partly hidden by trees and flood-darkened ground. Our models have the same difficulty.

3. Machine learning basics, as used here

3.1 Features vs labels

- A **feature** is a number that describes an example. Here, one example is one building, and a feature might be "how much rougher the roof became" (`gray_std_delta`). Each building gets **37 features**.
- The **label** is the answer we want to predict: the damage class. Labels come from the human annotations (the damage mask). We call them **ground truth**.

A **model** learns a rule that maps features to labels from many examples where both are known. It can then predict the label for a new building from its features alone.

3.2 Training, validation and test sets

We split the data into three parts:

- **Training set:** the model learns from it.
- **Validation set:** used to choose settings (**hyperparameters**, such as "how many neighbours should KNN use?") by comparing scores.
- **Test set:** used **once, at the end**, to measure how well the final model does on data it has never seen.

Think of it as homework (training), mock exams (validation) and the final exam (test). If you look at the final exam paper while studying, your mark no longer tells anyone how much you learned. In AfterSight a code object, `TestSetGuard` (`backend/app/evaluation/metrics.py`), raises an error if any model's test set is evaluated twice.

The splits are made per scene, not per building. Buildings in the same scene share the same photo, lighting and often the same damage. If neighbouring houses landed in both training and test, the test would be partly "seen before". So every building of a scene goes to the same split (Figure 3).

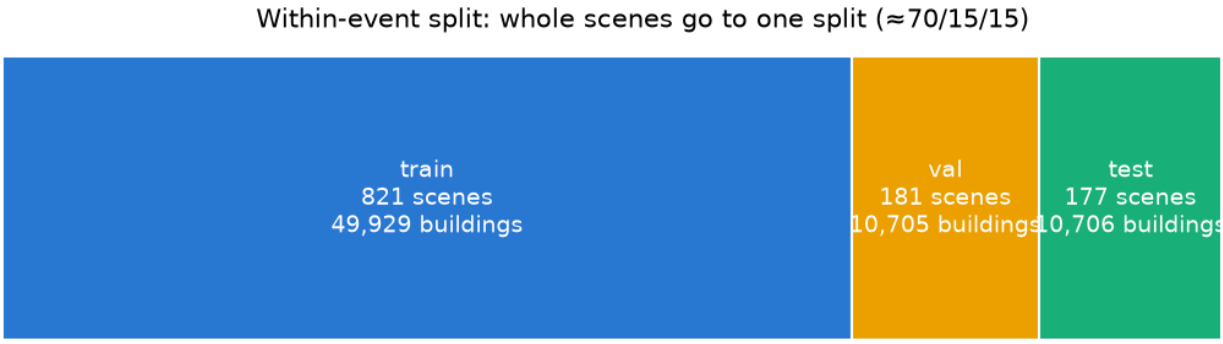


Figure 3: The within-event split (`data/splits/within_event_v1.json`): 821 training scenes (49,929 buildings), 181 validation scenes (10,705) and 177 test scenes (10,706).

There is a second split design, **cross-event**. It puts **every hurricane-michael scene that contains buildings (333 scenes, 21,429 buildings) in test**, and trains on the other four disasters. It answers "how well does this work on a disaster it has never seen?"

3.3 Classification vs clustering

- **Classification** learns from labelled examples to predict a label (supervised learning). Logistic Regression, KNN and Decision Trees are classifiers.
- **Clustering** groups examples by similarity **without labels** (unsupervised learning). K-Means is a clustering method. We use it only to ask whether buildings naturally fall into damage groups. Its groups

are never used as predictions.

3.4 Overfitting

A model **overfits** when it memorises the training data, noise included, instead of learning general patterns. It looks great on training data and worse on new data. Examples here:

- A Decision Tree that is allowed to grow very deep can create a leaf for almost every training building. We limit its depth (we tried 4, 6, 8 and 12; validation chose 8) and the minimum number of buildings per leaf (we tried 20 and 100; validation chose 100).
- KNN with k = 1 would copy the label of the single nearest building, noise included. We tried k = 5, 15 and 31; validation chose 15.

3.5 Class imbalance

58% of our buildings are "no damage" (41,449 of 71,340). A lazy model that always answered "no damage" would be right 58% of the time and completely useless. That is **class imbalance**. We handle it in two ways:

1. **Balanced class weights:** during training, mistakes on rare classes count more than mistakes on common ones.
2. **Macro-F1** as the main score: it gives each class equal importance (Section 7).

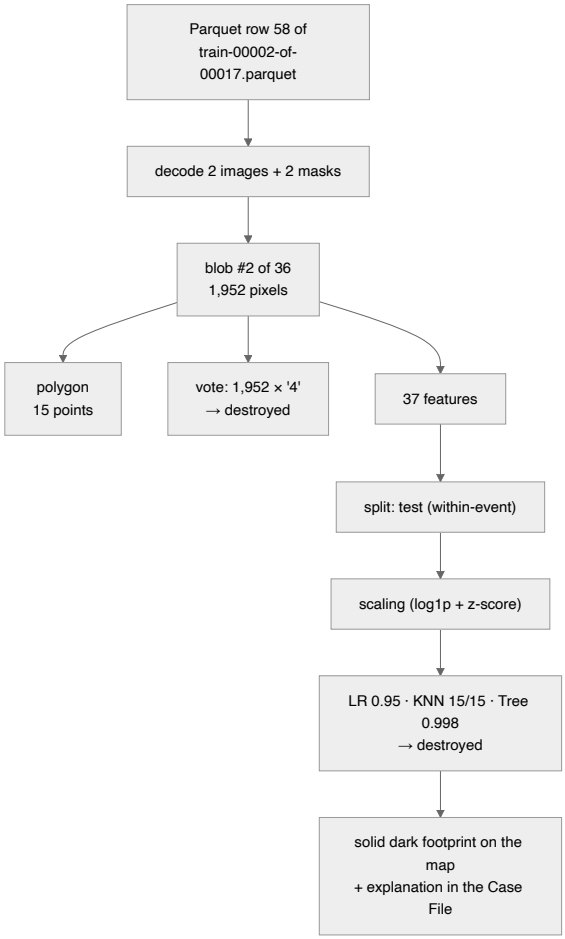
3.6 Why classical ML instead of deep learning

Deep learning (for example convolutional neural networks, CNNs) learns its own features from raw pixels. It is the state of the art on xBD, but it needs GPUs, is hard to explain, and is outside our course syllabus.

Classical ML uses features that *we* design and models that are simple enough to inspect. Every AfterSight prediction can be explained: a list of feature contributions, a path of yes/no questions, or the similar buildings it was compared with. The price is lower accuracy, and we are honest about it.

4. Follow one real building end to end

We follow `santa-rosa-wildfire_00000006_b0002`: a house in a held-out test scene of the 2017 Santa Rosa wildfire. It is one of the three demo scenes. Every number below was produced by `docs/build/understanding_data.py`, using the project's own code.



Step 1: the Parquet row

The scene is stored as **row 58** of `data/raw/xbd_parquet/train-00002-of-00017.parquet`. Its `image_name` is `train/images/santa-rosa-wildfire_00000006`, which is how we know the event and the scene number. The row holds four PNG files as raw bytes:

Column	What it is	Size
`t1_image`	pre-disaster photo	1,495,114 bytes
`t2_image`	post-disaster photo	1,593,407 bytes
`t1_mask`	footprint mask	8,814 bytes
`t2_mask`	damage mask	8,944 bytes

(Masks are tiny because they are mostly zeros, which compress very well.)

Code: `backend/app/datasets/xbd_parquet.py` reads rows a few at a time (`iter_rows`), so the 8 GB of files are never loaded at once.

Step 2: decoding the images and masks

Decoding the PNGs gives two **1024 × 1024 RGB** images and two masks. Counting the mask values:

- footprint mask: **902,934** pixels are 0 (background) and **145,642** are 1 (building), so about 14% of this scene is buildings;
- damage mask: 0 → 902,969; 1 → 65,338; 2 → 224; 3 → 53,432; **4 → 26,613** pixels.

Only the values 0–4 appear. A full scan of all 2,799 scenes confirmed this: there is no unexpected value.

At the centre of our building, the pre-image pixel is **(98, 103, 97)**, a dark grey roof. The post-image pixel is **(171, 168, 151)**, much brighter grey-white ash. That brightening is the first hint of what happened.

Step 3: finding the building blob

The footprint mask is just 0s and 1s. To find *separate* buildings, we look for **connected components** (blobs): groups of 1-pixels that touch along an edge. This is called 4-connectivity: left, right, up or down, not diagonally. `scipy.ndimage.label` numbers the blobs in reading order, top-left first. This scene has **36 blobs**, and all 36 were kept.

Our building is **blob number 2**, which is why its ID ends in `_b0002`. It spans rows 123–184 and columns 859–892, and contains **1,952 pixels**.

Code: `backend/app/datasets/mask_buildings.py` → `buildings_from_masks()`.

Step 4: turning the blob into a polygon

For drawing on screen we need an outline. OpenCV's `findContours` traces the blob's edge into a **polygon** with **15 points** (the first point is repeated at the end to close the shape), stored as text: `POLYGON ((872.5 123.5, 872.5 127.5, 871.5 128.5, 859.5 128.5, …))`. Coordinates are **image pixels** (x across, y down), not map coordinates.

The polygon's area is 1,859 px², slightly less than the 1,952-pixel blob, because the outline runs through the centres of the edge pixels. For measurements we always use the exact pixel count.

Step 5: the majority-vote label

We read the damage-mask values under the blob's 1,952 pixels. **All 1,952 are 4**, so the label is **destroyed**, with a **purity** of 1.0 (100% agreement). If two buildings had been touching, the blob might mix values (say 140 pixels of "3" and 60 of "2"). The majority would win and the blob would be logged as *mixed*. Ties, blobs with no damage value and blobs smaller than 16 pixels are dropped, and each drop is logged with its reason.



Figure 4: Our building (cyan outline) in all four layers. In the damage mask, the bright yellow means value 4 (destroyed).

Step 6: computing its 37 features

`backend/app/features/extract.py` measures the building's pixels in both images. A sample of the real values:

Feature	Value	What it means
`pre_mean_r` → `post_mean_r`	89.2 → 155.4	average red got much higher: the roof turned light grey
`abs_dmean_r`	66.3	size of that change in red
`gray_absdiff_mean`	60.1	on average, each pixel's brightness changed by 60 (out of 255)
`brightness_ratio`	1.64	post is 64% brighter than pre
`gray_std_pre` → `gray_std_post`	11.4 → 23.4	brightness spread inside the roof doubled: it became patchy
`gray_std_delta`	+12.0	that roughness change
`edge_density_pre` → `edge_density_post`	0.023 → 0.181	share of roof pixels on an edge (Canny detector) jumped from 2% to 18%: rubble has many edges
`glcm_homogeneity_pre` → `_post`	0.755 → 0.515	neighbouring pixels became much less alike (texture broke up)
`glcm_homogeneity_delta`	−0.240	that drop
`ring_dmean`	+32.1	the ground around the house also got brighter (ash everywhere)
`contrast_change`	+27.5	the house brightened *more* than its surroundings
`area_px`	1,952	size in pixels
`compactness`	0.678	how compact the shape is (1 = circle, 0.785 = square)

Table 2: Real feature values for `santa-rosa-wildfire_00000006_b0002`.

Two terms in the table: the **Canny detector** is a standard algorithm that marks pixels on sharp brightness edges, and **GLCM** (grey-level co-occurrence matrix) statistics describe texture by counting how often neighbouring pixels have similar brightness. Homogeneity is 1 when neighbours are identical.

Every sign points to destruction: brighter, rougher, more edges, less uniform. All 37 formulas are in `docs/FEATURES.md`.

Step 7: its split

In the **within-event** design this scene is a **test** scene: the models never trained on it. In the **cross-event** design it is a **training** scene, because only hurricane-michael is held out there. The app always shows which split a scene belongs to, and warns when a scene was used for training.

Step 8: scaling

Before the models see the features, the pipeline transforms them using statistics **learned from the training split only**:

1. **log1p** (log of 1 + x) on the 7 most skewed features, which have a very long tail of large values. For example, `area_px` 1,952 → log1p = 7.577.
2. **Standardise** (z-score): subtract the training mean and divide by the training standard deviation, so every feature is on a similar scale.

Feature	Raw	After log1p	Training mean	Training std	z-score
`gray_std_delta`	12.00	(not logged)	−0.92	10.91	**+1.18**
`glcm_homogeneity_delta`	−0.240	(not logged)	0.006	0.110	**−2.23**
`gray_absdiff_mean`	60.09	4.112	3.234	0.466	**+1.89**
`area_px`	1,952	7.577	6.783	1.032	**+0.77**

Table 3: Real scaling values. A z-score of −2.23 means "2.23 standard deviations below a typical training building", which is very unusual.

Step 9: how each model predicts

Logistic Regression (C = 10; C is a setting explained in Section 6.2). It computes a score for each class by multiplying every scaled feature by a learned weight (a **coefficient**) and adding them up, then turns the scores into probabilities. For *destroyed*, the biggest pushes were:

Feature	z-score	Coefficient	Contribution
`gray_std_delta`	+1.18	+5.57	**+6.60**
`dstd_g` (change in green spread)	+1.15	−2.74	−3.15
`post_mean_b` (mean blue, post)	+1.32	−2.28	−3.01
`dstd_r`	+1.17	−2.11	−2.48
`brightness_ratio`	+2.98	−0.76	−2.27
`ring_dmean`	+1.95	+1.04	+2.03

Negative contributions push *away* from destroyed, but they are outweighed: what matters is the total. The final *destroyed* score was **3.19** (including the intercept of −0.85 and all the other features). The probabilities came out as **destroyed 0.951**, minor 0.031, major 0.010, none 0.008.

KNN (k = 15). It finds the 15 training buildings whose 37 scaled features are closest to ours. **All 15 are destroyed buildings**, all from other Santa Rosa scenes (e.g. `santa-rosa-wildfire_00000007_b0047` at distance 1.45). Vote: 15 of 15 → destroyed, confidence 1.0.

Decision Tree (depth 8). It asks yes/no questions. Our building's path:

1. `glcm_homogeneity_delta` −0.240 ≤ −0.146? **yes**
2. `glcm_homogeneity_post` 0.515 ≤ 0.576? **yes**
3. `area_px` 1,952 > 521? **yes**
4. `glcm_homogeneity_delta` −0.240 ≤ −0.194? **yes**
5. `brightness_ratio` 1.64 > 0.98? **yes**
6. `post_mean_r` 155.4 > 127.8? **yes**
7. `ring_dmean` 32.1 > 5.7? **yes**
8. `contrast_change` 27.5 > −6.2? **yes** → a leaf of **971 training buildings**, almost all destroyed (0.998)

The optional **Random Forest** also says destroyed (0.997).

Step 10: the confidence

The **confidence** is the highest class score. What it means depends on the model:

- LR: a probability, though not *calibrated*, meaning nobody checked that predictions at 0.9 are right 90% of the time (0.951);

- KNN: the share of neighbours voting for the class (1.0);
- Tree: the class share in the leaf (0.998).

Each model has a **review threshold** chosen on validation: LR 0.484, KNN 0.533, Tree 0.526. Our building's confidence is well above all of them, so it is **not** flagged for human review.

Step 11: how it appears on the map

When you click **Run analysis** on the Triage Desk with LR, the API returns this building as `predicted: destroyed, confidence: 0.951, needs_review: false`, cached as analysis `068527454bebc6bf`. The browser draws its polygon over the satellite image with the *destroyed* style: the **darkest plum, solid fill**. Each class has its own pattern (outline, dots, hatch, solid), so you can tell them apart without relying on colour. In "Correct vs wrong" mode it gets a thin solid green outline, because the prediction matches the ground truth.

For the whole scene, LR got **31 of 36** buildings right (accuracy 0.861, scene macro-F1 0.735).

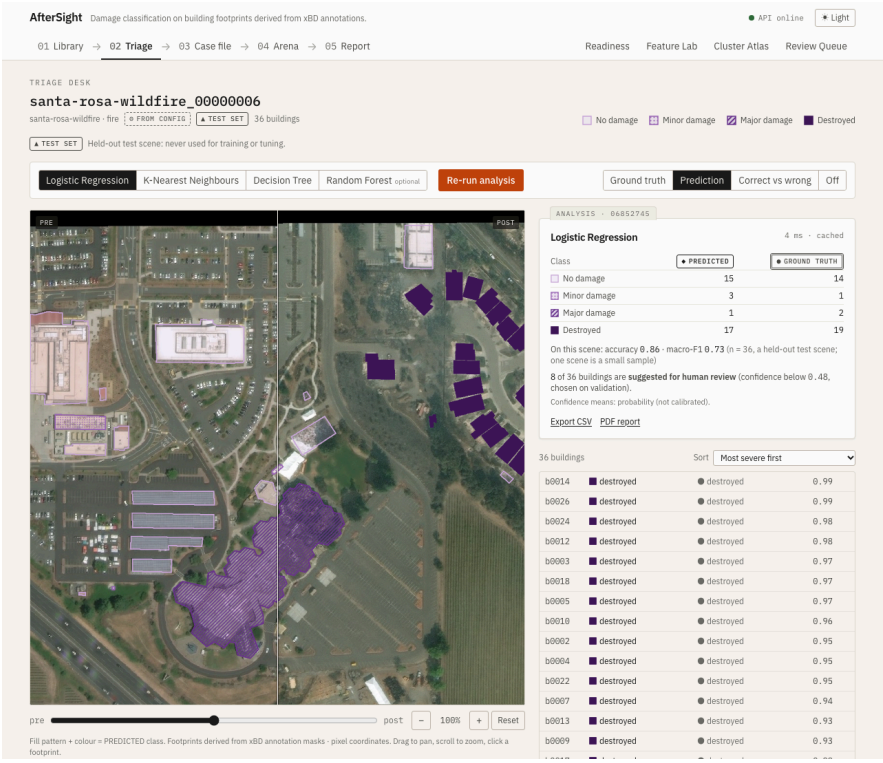


Figure 5: Triage Desk on `santa-rosa-wildfire_00000006` with LR predictions.

Step 12: how the Case File explains it

Clicking the footprint opens its **Case File**. It shows:

- the pre and post crops with the outline;
- *predicted: destroyed* vs *ground truth: destroyed* (✓ agrees);
- the class scores;
- the LR contribution bars from Step 9.

With KNN selected, it shows the 15 neighbour thumbnails instead; with the Tree, the 8 questions.

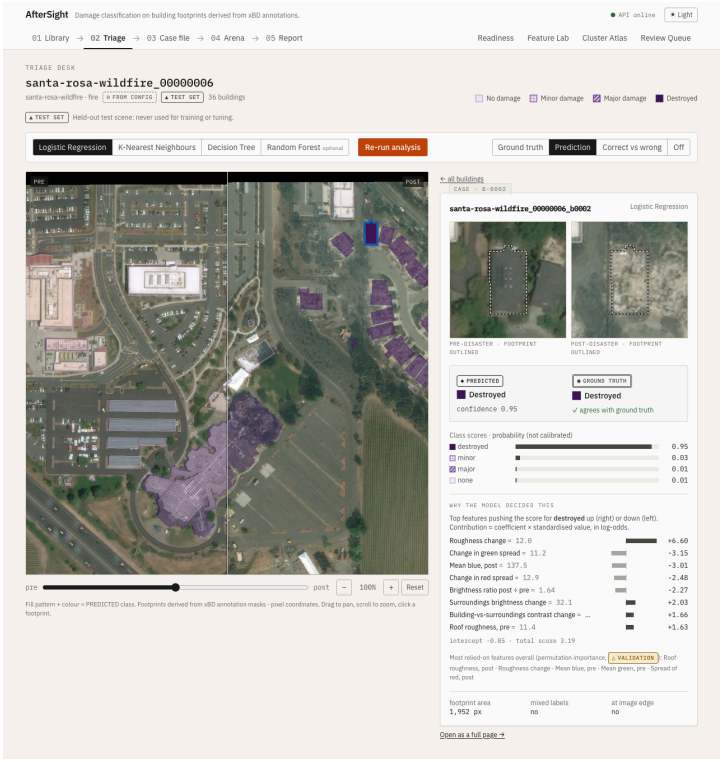
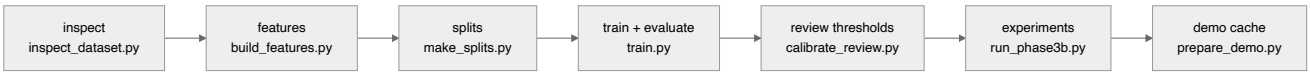


Figure 6: The real Case File for this building.

5. Each pipeline stage explained



`scripts/run_pipeline.sh` runs all of them in this order. A fresh run took 571 seconds on our laptop.

Loading (`backend/app/datasets/xbd_parquet.py`)

- **Does:** streams Parquet rows, decodes PNG bytes, and reads the event and scene from `image_name`.
- **Why:** the data is 8 GB, more than is sensible to hold in memory.
- **In → out:** 17 Parquet files → one row at a time (images and masks).
- **Mistake avoided:** running out of memory. Peak memory stayed around 1.2 GB.

Inspecting (`scripts/inspect_dataset.py`, `backend/app/datasets/inspection.py`)

- **Does:** checks every scene, counts classes per event, checks mask values, measures footprint sizes, and flags misaligned pre/post pairs.
- **Why:** before building on data, prove what is actually in it.
- **In → out:** shards → `artifacts/inspection/xbd/inspection_report.json`, CSV files and figures.
- **Mistake avoided:** assuming the data format. This is how we found that the mirror has masks, not polygons, and that two events had almost no damaged buildings.

Building extraction (`backend/app/datasets/mask_buildings.py`, `backend/app/datasets/scene_cache.py`)

- **Does:** connected components → buildings, majority-vote labels, polygons; caches the 5 selected events per scene.
- **Why:** the models need one example per building, and the app needs outlines.
- **In → out:** masks → `data/processed/scenes/<scene_id>/scene.json`, plus the images.
- **Mistake avoided:** silently giving a class to unlabelled buildings. 1,396 blobs with no damage value were excluded and counted instead.

Features (`backend/app/features/extract.py`, `backend/app/features/table.py`)

- **Does:** computes 37 numbers per building.
- **Why:** classical models need numbers, not images.
- **In → out:** scene cache → `data/processed/features/features_v1.parquet` (71,340 rows, 22.8 MB).
- **Mistake avoided:** feeding the label or the event into the model by accident. Identifiers are stored separately, and a test checks this.

Splits (`backend/app/training/splits.py`)

- **Does:** assigns whole scenes to train, validation or test, for both designs.
- **Why:** to get honest scores.
- **In → out:** feature table → `data/splits/within_event_v1.json` and `cross_event_v1.json`.
- **Mistake avoided:** leakage, i.e. the same scene in two splits. It is checked automatically before saving.

Training (`backend/app/training/run.py`, `backend/app/training/models.py`)

- **Does:** builds each pipeline (log1p → impute → scale → model), tries every setting in a small grid, keeps the best on validation, and saves it.
- **Why:** to find good settings without touching test.

- **In → out:** features + split → `artifacts/models/<design>/<scheme>/<model>.joblib` + metadata JSON.
- **Mistake avoided: data leakage through preprocessing**. Scaling statistics are learned from training data only, inside the pipeline.

Evaluation (`backend/app/evaluation/metrics.py`, `backend/app/evaluation/analysis.py`)

- **Does:** computes metrics on test once per model, plus per-event scores, the confounding check and K-Means.
- **Why:** to measure, and to explain the measurement.
- **In → out:** model + test data → `artifacts/metrics/…/comparison.json`, confusion-matrix figures.
- **Mistake avoided:** evaluating test repeatedly until a number looks good. `TestSetGuard` forbids it.

API (`backend/app/api/routes.py`, `backend/app/core/store.py`, `backend/app/inference/*`)

- **Does:** 25 web endpoints that serve scenes, images, analyses, explanations and reports from the saved files.
- **Why:** the browser cannot run Python models itself.
- **In → out:** HTTP requests → JSON, JPEG, CSV or PDF.
- **Mistake avoided:** letting a request read any file (path traversal). Only known IDs are accepted, never file paths.

Frontend (`frontend/src/pages/*`, `frontend/src/viewer/SceneViewer.tsx`)

- **Does:** the web pages.
- **Why:** to make the results explorable and understandable.
- **In → out:** API responses → screens.
- **Mistake avoided:** showing numbers that don't come from data. Every number on screen comes from an API response, and tests check this.

6. The algorithms, intuitively

6.1 The threshold baseline (not machine learning)

Idea. Take a single feature and cut it into four ranges with three thresholds. Each range is one class. It's like grading exams by one question: below 40 is "fail", 40–60 "pass", and so on.

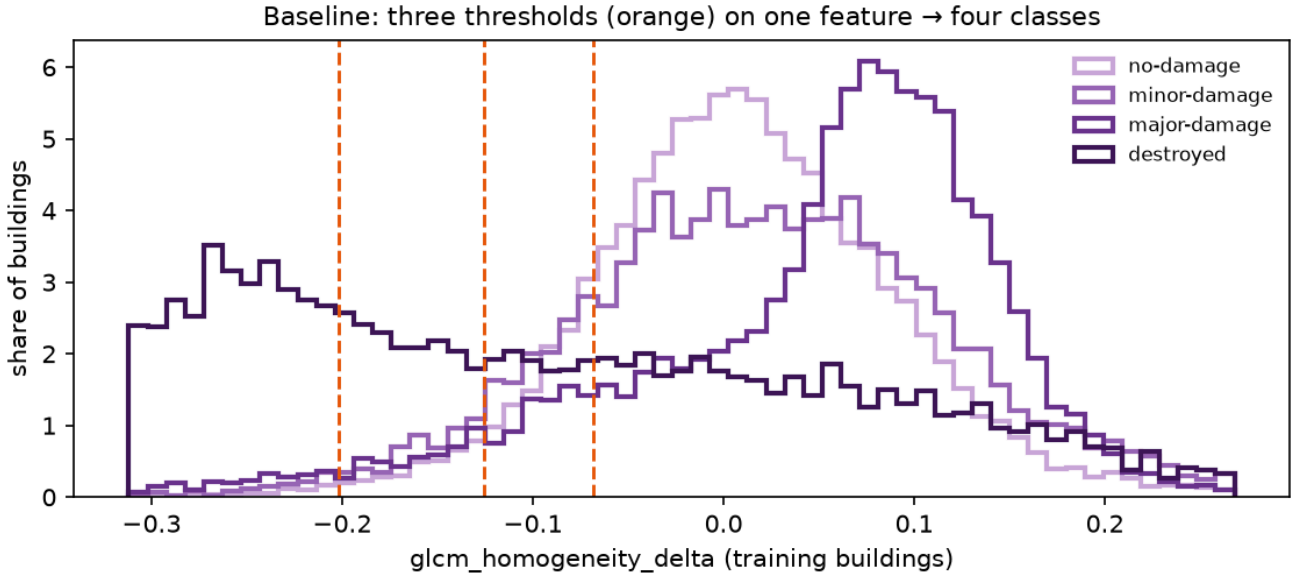


Figure 7: The chosen feature, `glcm_homogeneity_delta`, for training buildings of each class, with the three learned thresholds at −0.068, −0.126 and −0.202 (orange). The more negative the value, the more severe the predicted class.

Formula. Flip the sign so that larger means worse: v = − `glcm_homogeneity_delta`. With thresholds t_1 = 0.068 < t_2 = 0.126 < t_3 = 0.202, the class is the number of thresholds v passes: 0 = no damage, 1 = minor, 2 = major, 3 = destroyed. Our building has v = 0.240, which passes all three, so it is predicted *destroyed*.

- **Pros:** trivially explainable; a floor that any real model must beat.
- **Cons:** one feature cannot capture everything. Major damage actually *peaks at positive values* in Figure 7, so the baseline misses it.
- **Here:** test macro-F1 0.383 within-event and 0.193 cross-event.

6.2 Logistic Regression

Idea. Give each feature a weight. Multiply, add up, and the class with the highest total wins. Then squash the totals into probabilities. **Analogy:** a points checklist, where "+5.6 points for each step of extra roughness, −2.3 for each step of extra brightness ratio…".

Formula. For each class *k*: score `z_k = w_k · x + b_k`, then probability `p_k = e^{z_k} / Σ_j e^{z_j}` (the **softmax** function).

- *x* is the vector of 37 scaled features.
- *w_k* holds the 37 learned weights for class *k*.
- *b_k* is the intercept (bias).
- *e* is Euler's number (≈ 2.718).
- The Σ adds up over all four classes, so the probabilities sum to 1.

With two classes this reduces to the familiar **sigmoid** `1 / (1 + e^{−z})`.

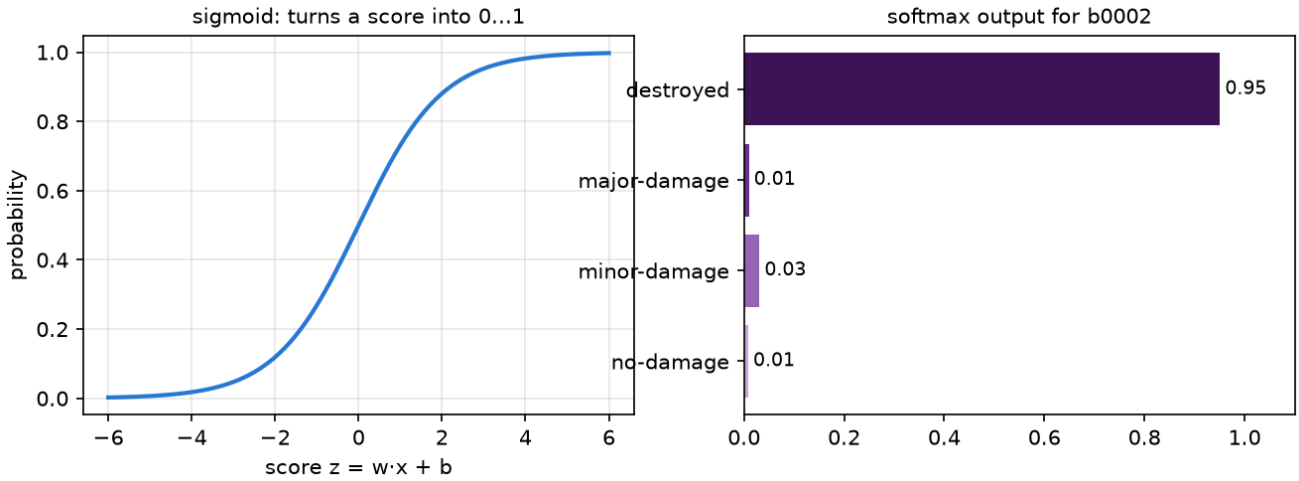


Figure 8: Left, the sigmoid turns any score into a number between 0 and 1. Right, LR's real output for our building.

C controls **regularisation**: how strongly the weights are kept small to avoid overfitting. Smaller C means stronger regularisation. We tried 0.01, 0.1, 1 and 10; validation chose 10.

- **Pros:** fast (training took 1.1 s); weights are directly interpretable; outputs probability-like scores.
- **Cons:** assumes each feature pushes in one direction (linear), so it cannot learn "medium is bad, but low and high are fine".
- **Here:** within-event test macro-F1 **0.527**, cross-event **0.290**. It is the app's default model.

6.3 K-Nearest Neighbours (KNN)

Idea. "Show me the most similar buildings I've already seen, and copy their majority label." **Analogy:** guessing a house price from the prices of the most similar houses nearby.



Figure 9: Training buildings plotted on two of the 37 features. Our building is the star; its 15 real neighbours (found using all 37 features) are circled.

Formula. Distance between buildings a and b: $d(a, b) = \sqrt{\sum_i (a_i - b_i)^2}$ (Euclidean distance over the 37 scaled features i). Pick the k smallest distances; the predicted class is the most common label among them; confidence = votes for that class ÷ k.

- **Pros:** no training needed; the explanation is concrete ("these 15 similar buildings were destroyed").
- **Cons:** slow at prediction time, because it compares with all 49,929 training buildings (about 20 ms per 1,000 buildings, against 0.2 ms for LR). Sensitive to how features are scaled.
- **Here:** k = 15 with uniform votes; test macro-F1 0.520 within-event, 0.271 cross-event.

6.4 Decision Tree

Idea. A flowchart of yes/no questions, learned from data. **Analogy:** a doctor's triage checklist ("fever above 38? → yes → rash? → ...").

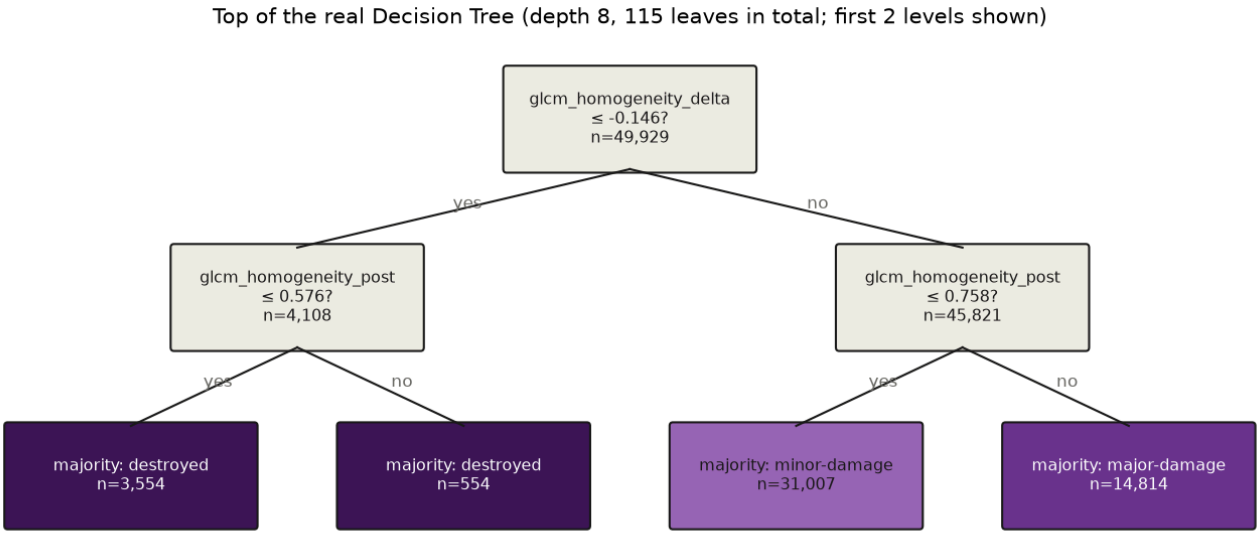


Figure 10: The first two levels of the real tree. The very first question is about the texture change, the most informative single split.

How a question is chosen. At each node, the tree tries many feature/threshold pairs and keeps the one that makes the two groups most "pure", using **Gini impurity**: $G = 1 - \sum_k p_k^2$, where p_k is the share of class k in the group. G = 0 means every building in the group has the same class.

- **Pros:** completely readable; handles non-linear patterns; no scaling needed in principle.
- **Cons:** a single tree is unstable (small data changes can change the tree) and overfits if grown too deep.
- **Here:** depth 8, at least 100 buildings per leaf, 115 leaves; test macro-F1 0.470 within-event, 0.294 cross-event (the best cross-event score, by a hair).

Random Forest (optional) averages 200 such trees, each grown on a random sample of data and features. It is more accurate (0.546 within-event) but loses the single readable path.

6.5 K-Means (clustering, analysis only)

Idea. Choose k, place k centre points, assign every building to its nearest centre, move each centre to the average of its buildings, and repeat until nothing changes. **Analogy:** sorting a pile of socks into k heaps by how alike they look, without knowing which pairs belong together.

Formula. K-Means minimises the **inertia** $\sum_{buildings} \|x - \mu_{c(x)}\|^2$: the total squared distance from each building x to the centre μ of its cluster c. The **silhouette** score (–1 to 1) measures how much closer each point is to its own cluster than to the next one.

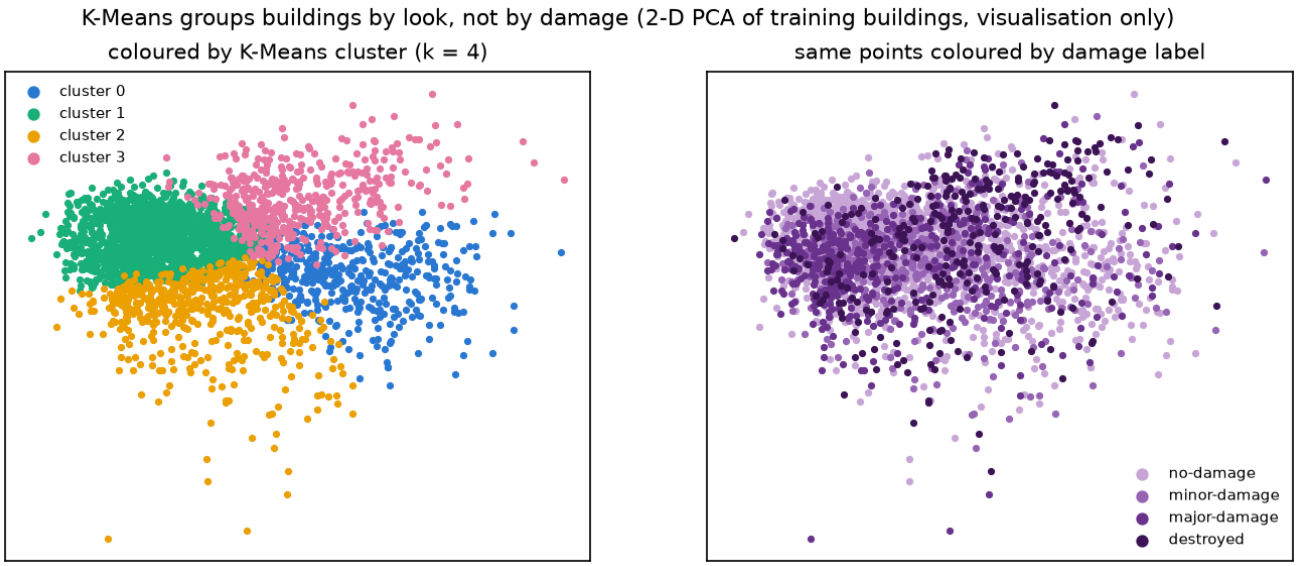


Figure 11: The same training buildings coloured by K-Means cluster (left) and by true damage (right). The groups do not line up.

- **Pros:** simple; shows the natural structure of the data.
- **Cons:** you must choose *k*; clusters have no meaning attached.
- **Here:** the best silhouette is at k = 2 (0.253). Agreement with damage labels is weak: adjusted Rand index 0.041 at k = 2 and 0.095 at k = 4.

7. Understanding the results

7.1 Reading a confusion matrix

A **confusion matrix** counts, for every true class (row), how often the model predicted each class (column). The diagonal holds the correct predictions. Here is LR on the within-event test set (10,706 buildings):

true ↓ / predicted →	no damage	minor	major	destroyed	total
no damage	**3,264**	779	1,757	420	6,220
minor	363	**815**	297	293	1,768
major	276	243	**871**	150	1,540
destroyed	61	90	65	**962**	1,178

Table 4: Real confusion matrix (`artifacts/models/within_event/four_class/logistic_regression.json`).

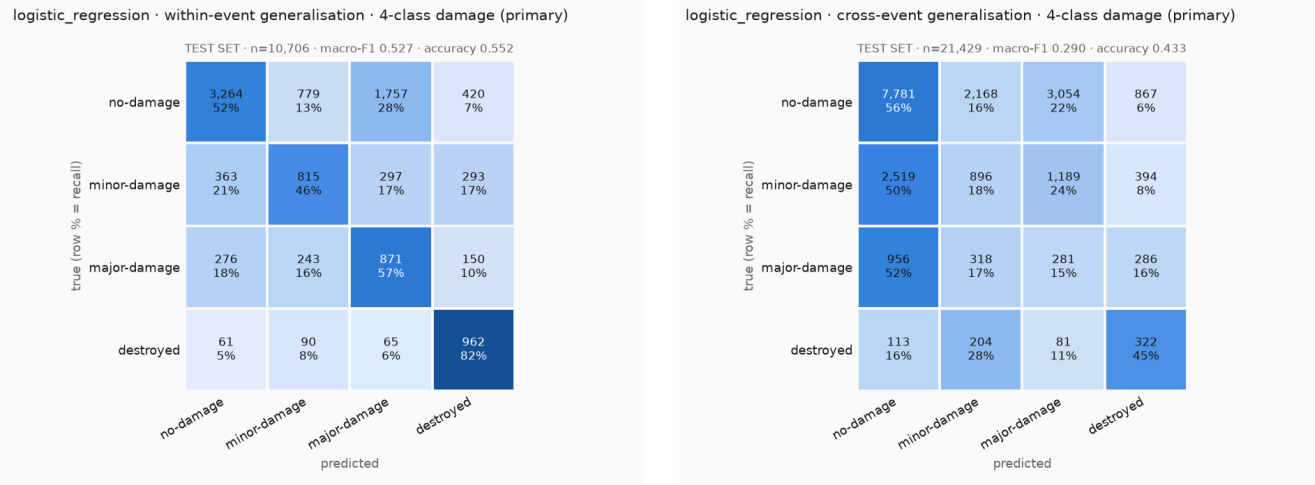


Figure 12: The same matrix as a heatmap (left) and the cross-event one (right). Row % = recall.

7.2 The metrics, with worked examples from Table 4

- **Accuracy** = correct / all = (3,264 + 815 + 871 + 962) / 10,706 = 5,912 / 10,706 = **0.552**.
- **Precision** for a class: of everything *predicted* as that class, the share that really was. For *destroyed*, the column total is 420 + 293 + 150 + 962 = 1,825, so precision = 962 / 1,825 = **0.527**.
- **Recall** for a class: of everything that really *is* that class, the share we caught. For *destroyed*: 962 / 1,178 = **0.817**.
- **F1** combines the two (harmonic mean): F1 = 2 · P · R / (P + R) = 2 · 0.527 · 0.817 / (0.527 + 0.817) = **0.641** for *destroyed*.
- **Macro-F1** is the plain average of the four per-class F1 scores: (0.641 + 0.441 + 0.385 + 0.641) / 4 = **0.527**.

Why macro-F1? The lazy "always no damage" model would have accuracy 6,220 / 10,706 = 0.58, *higher* than our model, but F1 = 0 for three classes, so macro-F1 ≈ 0.18. Macro-F1 cannot be fooled by imbalance.

Two things the matrix shows:

- **1,757 undamaged buildings were predicted as major damage.** This is the cost of balanced class weights, which make the model eager to find rare classes. It is why major damage has a low precision (0.291).
- **Destroyed is caught well** (recall 0.817), because rubble changes the texture a lot.

7.3 Why 0.53 within-event but 0.29 cross-event

Model	Within-event macro-F1	Cross-event macro-F1
Threshold baseline	0.383	0.193
Logistic Regression	0.527	0.290
KNN	0.520	0.271
Decision Tree	0.470	0.294
Random Forest (optional)	0.546	0.291

Table 5: Test results, each model evaluated once (`artifacts/metrics/*/four_class/comparison.json`).

Within-event, the test scenes come from the *same five disasters* as the training scenes: the same satellite, season, region and roof styles. Cross-event, hurricane-michael looks different, and its damage mix is different too (23% minor damage vs 14% in the training events). Every model loses between a third and a half of its score. **0.53 is an optimistic upper bound; 0.29 is the realistic figure for a new disaster.**

7.4 What the confounding check and Phase 3b proved

Confounding check. We trained a Logistic Regression to predict the **event** (which disaster) from the same 37 features. It was right **82.9%** of the time, against 25.2% for always guessing the most common event. So the features carry a strong "which disaster is this?" signature. Per-event scores confirm the effect: within-event, each disaster alone scores *below* the pooled 0.527 (harvey 0.333, matthew 0.333, michael 0.363, palu 0.529, santa-rosa 0.460). Part of the pooled score comes from recognising the disaster and its typical damage mix.

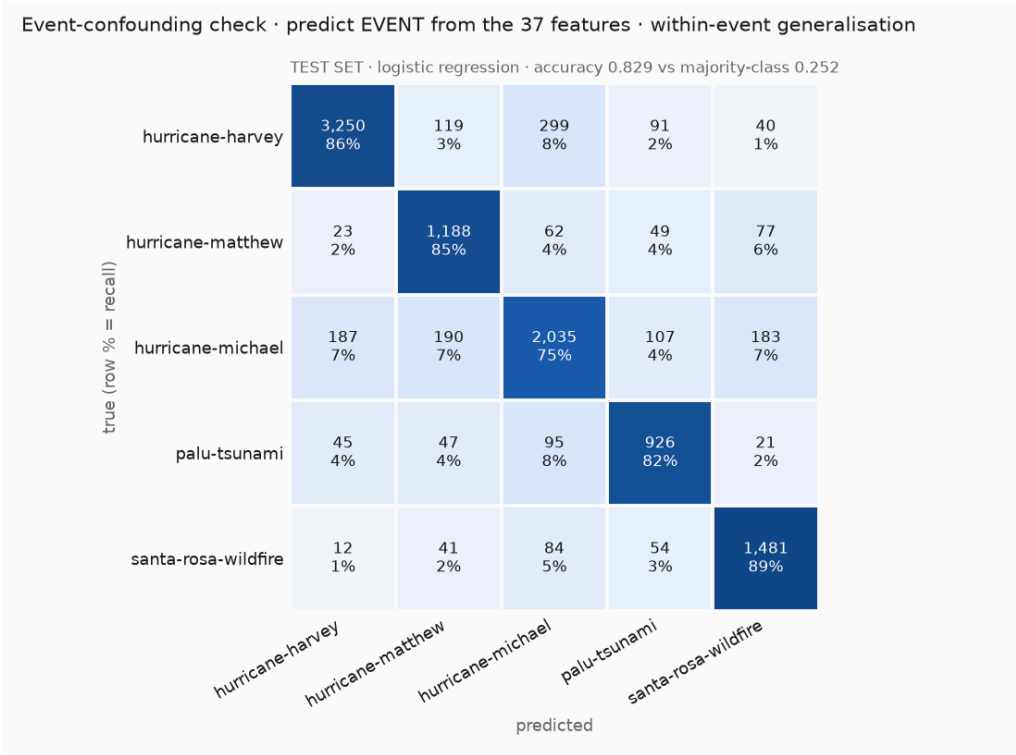


Figure 13: Predicting the event from the features (test set): most buildings are recognised.

Phase 3b. We tried to remove that signature:

- **change-only:** 17 features, with no absolute colour levels;
- **scene-normalised:** each feature minus the median of its scene;
- **both combined.**

Event prediction fell from 82.9% to as low as **37.4%**, so the signature shrank. But cross-event macro-F1 **did not improve** (0.236–0.278, against 0.290), and within-event got worse. **Conclusion:** the poor transfer is not mainly the model "cheating" with the event. It is a genuine **domain shift**: damage looks different in different disasters, and these hand-made features can't bridge that.

7.5 Why K-Means agrees weakly with damage

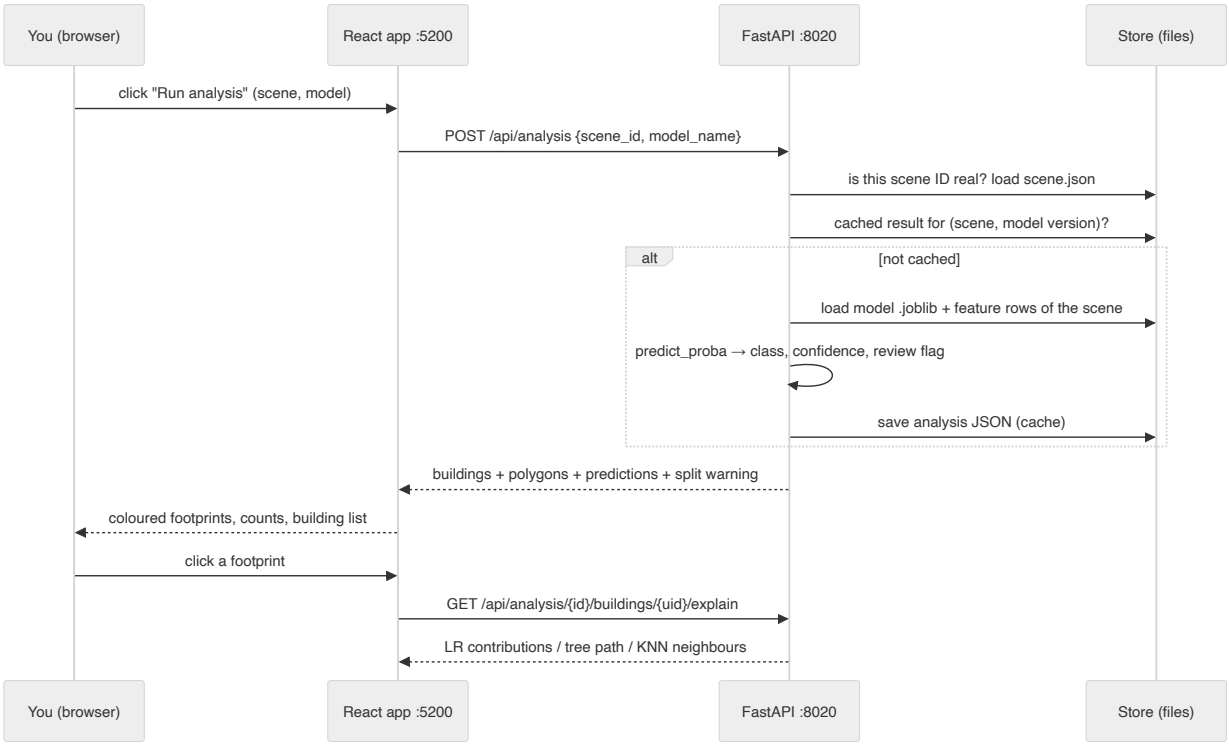
K-Means groups buildings by whatever differs most in the features: roof colour, vegetation, scene brightness, which disaster. Damage is only one of many sources of variation, and often not the biggest. The adjusted Rand index (0 = random agreement, 1 = perfect) is only **0.041–0.095**. That doesn't mean the features are useless: the supervised models *do* find damage, because they are told what to look for.

8. How the app works

8.1 Frontend ↔ API ↔ saved models

- The **frontend** (React, in your browser) only displays things.
- The **API** (FastAPI, Python, port 8020) loads saved models and data files and answers requests.
- Nothing is trained by the app. Training happens only when you run the pipeline scripts.

8.2 What happens when you click "Run analysis"



On a typical 40-building test scene this takes **8–22 ms** the first time, and about **3 ms** once cached.

8.3 What each page shows and why

Page	Shows	Why it exists
Readiness	what is built and trained; the 0.527 vs 0.290 headline; served models with their test macro-F1, confidence meaning and review threshold	know at a glance whether the system is ready, and how good it honestly is
Library	scenes, held-out **test** scenes by default, with ground-truth damage bars	choose a scene; training scenes carry a warning
Triage Desk	swipe between pre and post, zoom, overlays (ground truth / prediction / correct vs wrong), building list	the main workspace
Case File	one building: crops, predicted vs true, confidence, explanation	understand a single decision
Arena	every model, both designs, confusion matrices, per-event scores, tree, confounding, experiments	compare models honestly
Feature Lab	which features separate the classes (training split)	see *why* the models work or fail

Page	Shows	Why it exists
Cluster Atlas	precomputed K-Means for k = 2…10: cluster sizes, silhouette, ARI and which damage classes fall into each cluster; clearly marked "never predictions"	unsupervised view of the data; a cluster ID is not a damage class
Review Queue	buildings *suggested for human review* (low confidence or models disagree)	where a person should look first
Reports	CSV and PDF per analysis	take the results away

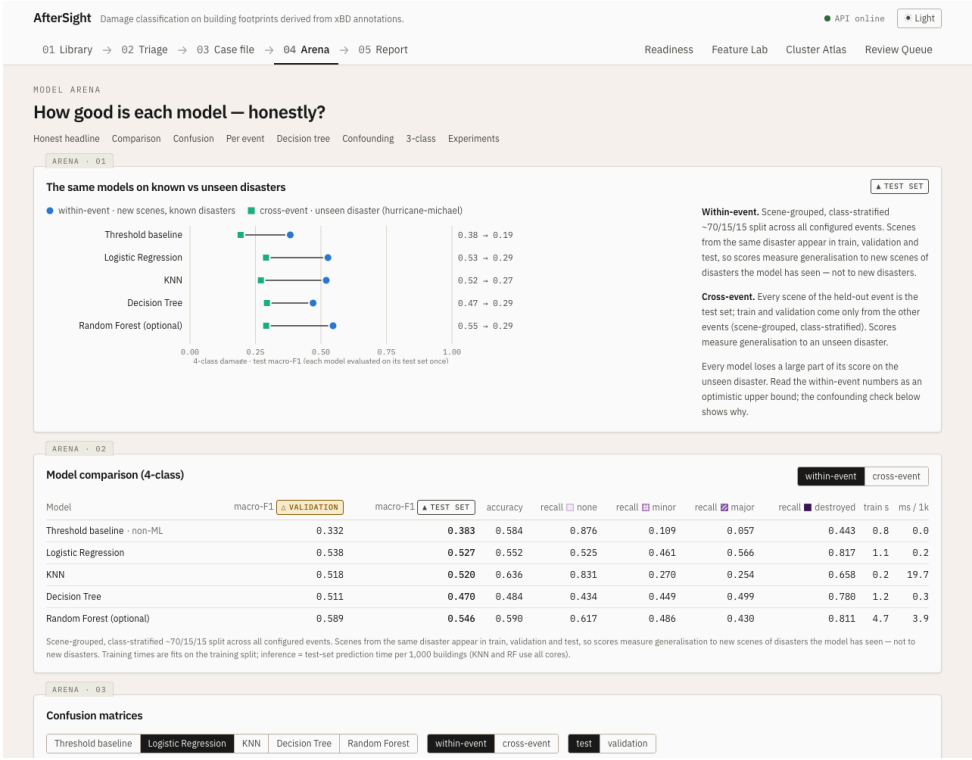


Figure 14: The Model Arena's first view: every model drops from within-event to cross-event.

9. Honest limitations

Limitation	Why it exists
Buildings come from **masks**, not original outlines; touching buildings merge into one blob	our copy of xBD (the Parquet mirror) has no polygons; 782 blobs mixed damage values
No real-world coordinates or areas	the mirror has no georeferencing, so polygons are in pixels only
Minor vs major are often confused	the difference is subtle from above at sub-metre resolution (Figure 2)
0.29 on a new disaster	domain shift between disasters (Section 7.4)
Confidence is not a calibrated probability	the models' scores were never calibrated; we only use them with validation thresholds
Some buildings (1,396 blobs) excluded	they had no damage value in the mask, and we refuse to guess a class
Only xBD's training split	the mirror contains only that split; all our test sets are carved out of it
Not for real rescue decisions	it is a research and teaching project with known error rates

10. Code map

Path	What it does
`scripts/run_pipeline.sh`	runs every build step in order
`start.sh`	starts the API (8020) and web app (5200), freeing the ports first
`scripts/inspect_dataset.py`	inspects the shards, fills the scene cache
`scripts/build_features.py`	builds the 37-feature table
`scripts/make_splits.py`	writes the two split manifests
`scripts/train.py`	trains, tunes and evaluates every model; K-Means; confounding check
`scripts/calibrate_review.py`	chooses review thresholds on validation
`scripts/run_phase3b.py`	the feature-variant experiments
`scripts/prepare_demo.py`	picks and pre-caches the 3 demo scenes
`scripts/e2e_check.py`	end-to-end test on synthetic data
`scripts/snapshot_behaviour.py`	proves a refactor changed no prediction, explanation or Feature Lab number
`config/events.yaml`	which events, disaster types, mask settings
`config/features.yaml`	feature settings (ring width, Canny, GLCM)
`config/training.yaml`	splits, model grids, serving models, review target
`backend/app/datasets/xbd_parquet.py`	streams the Parquet shards
`backend/app/datasets/mask_buildings.py`	masks → buildings, labels, polygons
`backend/app/datasets/inspection.py`	the dataset inspector
`backend/app/datasets/scene_cache.py`	per-scene cache on disk
`backend/app/features/extract.py`	computes the features for one building
`backend/app/features/schema.py`	names of the 37 features and the non-feature columns
`backend/app/features/table.py`	builds and loads the feature table
`backend/app/features/variants.py`	Phase 3b feature variants
`backend/app/training/splits.py`	scene-grouped splits and leakage checks
`backend/app/training/models.py`	pipelines, baseline rule, log1p transformer
`backend/app/training/run.py`	tuning, single test evaluation, saving
`backend/app/training/persistence.py`	save/load models with compatibility checks
`backend/app/evaluation/metrics.py`	metrics and `TestSetGuard`
`backend/app/evaluation/analysis.py`	K-Means and the confounding check
`backend/app/explain/explain.py`	LR contributions, tree paths, KNN neighbours
`backend/app/api/routes.py`	all API endpoints
`backend/app/core/store.py`	file-based data access, ID validation

Path	What it does
`backend/app/inference/analysis.py`	runs and caches an analysis; explanations
`backend/app/inference/review.py`	review thresholds: calibration on validation, which thresholds are usable, READY / PARTIAL / NOT READY
`backend/app/inference/review_queue.py`	builds the review queue
`backend/app/experiments/summary.py`	reads and checks the saved Phase 3b summary for the Model Arena
`backend/app/inference/feature_lab.py`	Feature Lab statistics (η^2, per-class percentiles, scatter sample), training split only
`backend/app/reports/reports.py`	CSV and PDF reports
`frontend/src/pages/`	one file per page (Triage Desk, Arena, …)
`frontend/src/viewer/SceneViewer.tsx`	the swipe/zoom/overlay viewer
`frontend/src/components/CaseFile.tsx`	the Building Case File
`frontend/src/charts/`	hand-made SVG charts
`frontend/DESIGN.md`	the visual design system
`docs/FEATURES.md`, `docs/RESULTS.md`, `docs/ML_EXPLAINED.md`, `docs/API.md`, `docs/VIVA_NOTES.md`	detailed references
`docs/SIMPLIFICATION_NOTES.md`	what the v1.1 simplification changed and how behaviour was kept identical
`docs/FIXES_V1_2.md`	the v1.2 status, wording and missing-file fixes
`docs/CHANGES_V1_3.md`	the v1.3 audit, environment diagnosis and Cluster Atlas changes
`backend/tests/`, `frontend/src/**/*.test.tsx`	148 backend and 37 frontend tests

11. Glossary

Term	Simple meaning
Adjusted Rand index (ARI)	how similar two groupings are: 1 = identical, 0 = no better than random
API	a set of web addresses a program can call to get data (here: FastAPI on port 8020)
Balanced class weights	making mistakes on rare classes count more during training
Baseline	the simplest reasonable method, used to show that a real model adds value
Blob / connected component	a group of touching pixels; here, one building
Canny edge detector	an algorithm that marks pixels on sharp brightness edges
Channel (RGB)	one colour layer of an image: red, green or blue
Classification	predicting a category from features, learned from labelled examples
Clustering	grouping examples by similarity without labels
Coefficient (weight)	how much a feature pushes a linear model's score up or down
Confidence	the model's highest class score; its meaning differs per model
Confounding	when something else (here, which disaster) is mixed up with what you want to measure
Confusion matrix	a table of true vs predicted classes
Cross-event	testing on a disaster never seen in training
Domain shift	when test data looks systematically different from training data
Feature	a number describing an example
F1 / macro-F1	combines precision and recall; macro = average over classes, each equal
GLCM	grey-level co-occurrence matrix: statistics of how neighbouring pixel brightnesses relate (texture)
Ground truth	the reference labels from human annotation
Hyperparameter	a setting chosen before training (C, k, depth), tuned on validation
Inertia	K-Means' total squared distance from points to their cluster centres
Label	the answer to predict (damage class)
Leakage	information from test data sneaking into training, which inflates scores
log1p	log(1 + x); shrinks very large values so a long tail matters less
Mask	an image whose pixels hold labels instead of colours
Overfitting	memorising training data instead of learning general patterns
Parquet	a compressed, column-based table file format
Pipeline	a fixed chain of preprocessing steps plus a model, fitted together
Polygon	an outline made of points; here in image pixel coordinates
Precision	of everything predicted as a class, the share that was right
Purity	the share of a blob's pixels that agree with its majority label

Term	Simple meaning
Recall	of everything that is a class, the share we found
Regularisation	keeping a model simple (e.g. small weights) to avoid overfitting
Resolution	how much ground one pixel covers (xBD: finer than 0.8 m)
Scene	one 1024 × 1024 pre/post image pair
Silhouette	how well each point fits its own cluster vs the next (–1 to 1)
Softmax	turns class scores into probabilities that sum to 1
Standardise (z-score)	(value – mean) / standard deviation, using training statistics
Test / validation / training set	final exam / mock exams / homework
Within-event	testing on new scenes from the same disasters as training
xBD / xView2	the public building-damage dataset and the challenge it was made for

12. Test yourself

1. What are the four damage classes?
2. What does one pixel of an RGB image store?
3. What is the difference between the footprint mask and the damage mask?
4. Why do we split by scene instead of by building?
5. What is the difference between validation and test data?
6. Our building had 1,952 pixels, all with damage value 4. What is its label, and what is its purity?
7. What does `brightness_ratio = 1.64` mean for our building?
8. Why is `log1p` applied to `area_px`?
9. Why is the scaler fitted only on training data?
10. Why is accuracy misleading on this dataset? Give the number.
11. From Table 4, compute recall for minor damage.
12. Why does major damage have low precision for LR?
13. How does KNN produce a confidence value? What was it for our building?
14. Name one pro and one con of a Decision Tree.
15. What does C control in Logistic Regression, and which value was chosen?
16. Why is K-Means not used for predictions?
17. What does the cross-event design hold out, and why?
18. The event can be predicted from the features with 82.9% accuracy. Why does that matter?
19. Phase 3b reduced event prediction to 37.4%. Why didn't that fix cross-event performance?
20. A building appears with a dashed red outline in "Correct vs wrong" mode. What does that mean, and what would you check in its Case File?

Answers

1. No damage, minor damage, major damage, destroyed.
2. Three numbers from 0 to 255: red, green and blue intensity.
3. The footprint mask marks *where* buildings are (0/1). The damage mask gives each building pixel a damage value (1–4).
4. Buildings in one scene share imagery and damage. Splitting them across train and test leaks information and inflates scores.
5. Validation is used to choose settings; test is used once at the end to measure the final model.
6. Destroyed, with purity 1.0 (all pixels agree).
7. After the disaster its pixels are on average 64% brighter than before (ash and rubble).
8. Area has a very long tail (up to 648,034 px). The log shrinks huge values so they don't dominate.
9. Otherwise information from validation and test would leak into training.
10. 58% of buildings are undamaged, so always answering "no damage" gives about 0.58 accuracy while being useless.
11. 815 / 1,768 = 0.461.
12. Balanced weights make it predict major often, and 1,757 undamaged buildings were wrongly predicted as major, so only 871 / 2,990 = 0.291 of "major" predictions were right.
13. The share of its *k* neighbours voting for the predicted class: 15 / 15 = 1.0.
14. Pro: readable yes/no rules. Con: unstable and overfits when grown deep.
15. Regularisation strength (smaller C = stronger). Validation chose C = 10.
16. Clusters are formed without labels and don't match damage (ARI 0.04–0.10); they have no damage meaning.
17. All of hurricane-michael (21,429 buildings), to measure performance on a disaster never seen in training.
18. Part of the within-event score may come from recognising the disaster, so 0.53 is optimistic.
19. The difficulty is genuine domain shift: damage looks different across disasters. Removing the signature also removed useful information.
20. The prediction disagrees with the ground truth. Check the explanation (which features or questions drove it), the confidence, whether it is flagged for review, and whether the footprint is a merged blob.

References

[1] R. Gupta *et al.*, "xBD: A Dataset for Assessing Building Damage from Satellite Imagery," arXiv:1911.09296, 2019. Other references (Haralick's GLCM, Canny, scikit-learn, Breiman, Rousseeuw, Hubert & Arabie) are listed in `report/AfterSight_Mini_Project_Report.pdf`.